OOAD – Activity Diagrams

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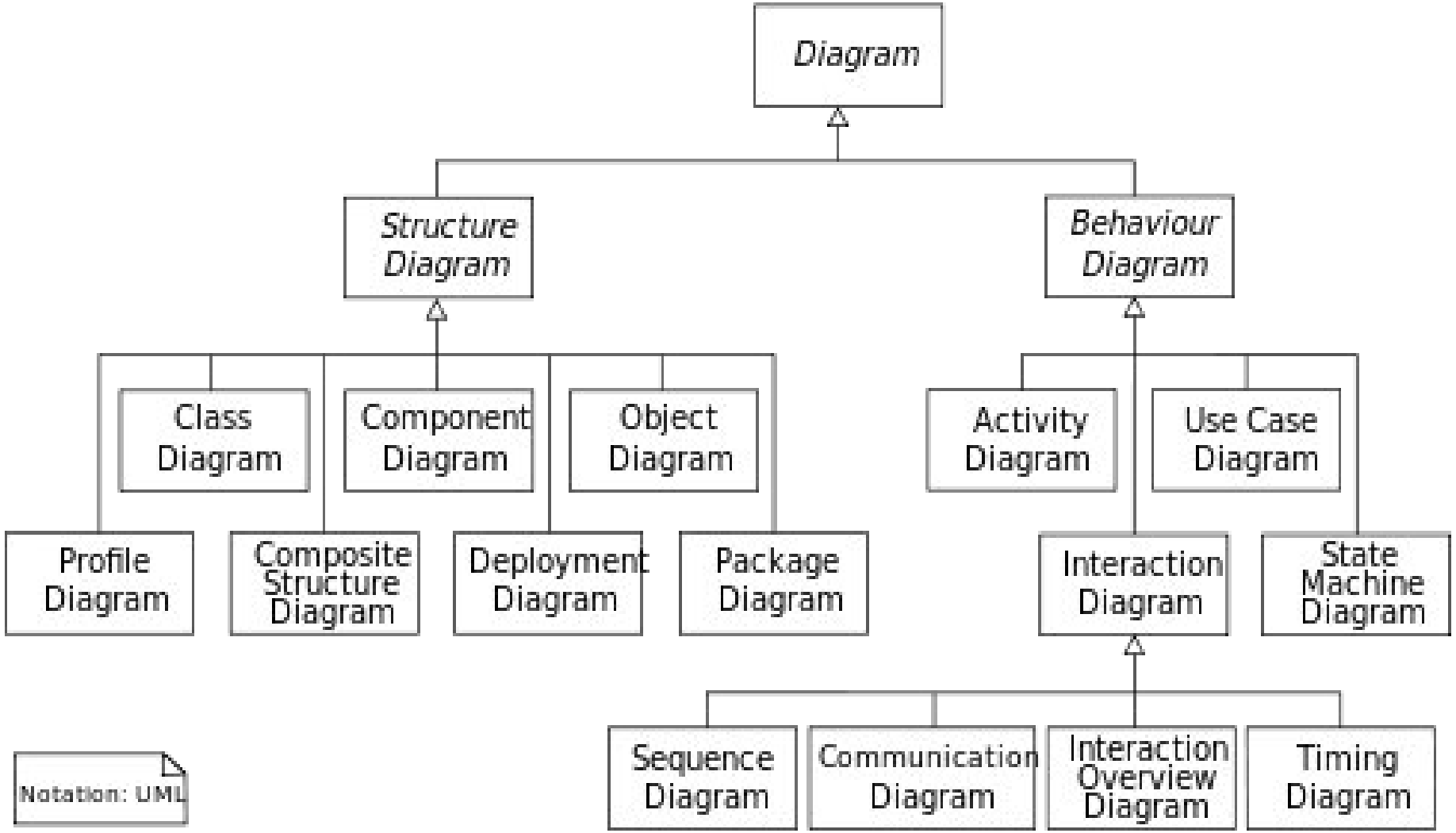
UML Diagrams



Business Process Modelling with Activity Diagrams – **Learning Outcomes**

- Model Use-Case activities using Activity Diagrams
- Identify and draw *swimlanes*, *synchronization bars* in activity diagrams
- Draw an Activity Diagram for a given Scenario

Working with Activity Diagrams

- Models the process steps or Use Case activities of the system.
- They represents the dynamics of a system.
- They are similar to flow charts .
- They graphically show the work flow of a system.

Working with Activity Diagrams

- They show the flow of control from activity to activity in the system.
- They show what activities can be done in parallel, (That is where they are different from flowcharts.)
- They also show alternative paths through the flow.

Working with Activity Diagrams cont..

- After creating a use case diagram, *activity diagrams* may be created to represent the flow ***across use cases*** or they may be created to represent the ***flow within a particular use case***.
- Activity diagrams may also be created to show the ***workflow for an operation.***

Working with Activity Diagrams cont..

Activity diagrams contain :

- Activities,
- Transitions between activities,
- Decision points,
- and synchronization bars.
- Swimlanes

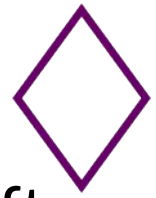
Activity

- Represented in UML by a rounded rectangle.
- Activity represents the performance of some behavior in the work flow.

Transitions

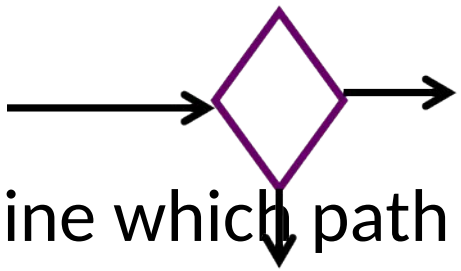
- Transitions are used to show the passing of the flow of control from activity to activity.
- They are typically triggered by the completion of the behavior in the originating activity.

Decision Points



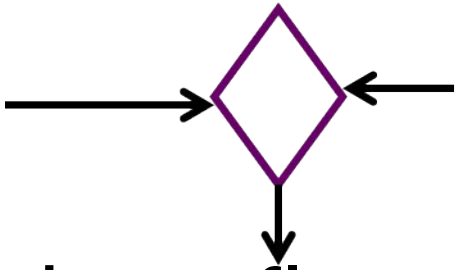
- When modeling the workflow of a system, it is often necessary to show where the flow of control branches based on a decision point.
- The transition from a decision point contain a guard condition.

Decision Points cont..



- The guard condition is used to determine which path from the decision point is taken.
- Decisions along with their guard conditions allow you to show alternative paths through a work flow.

Merge Point

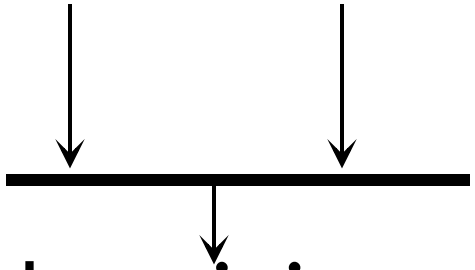


- Two or more flows come in and one flow goes out.
- This combines flows that were previously separated by decisions.
- Processing continues with any one flow coming into the merge.

Synchronization Bars

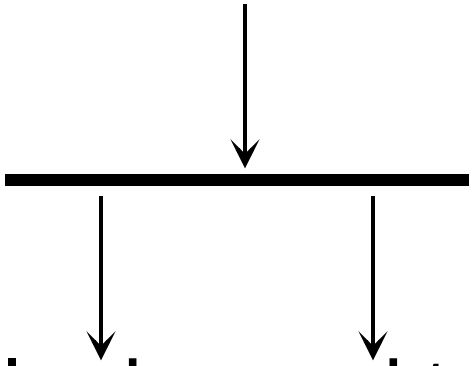
- In a workflow there are typically some activities that may be done in parallel.
- A synchronization bar allows you to specify what activities may be done concurrently.

Join



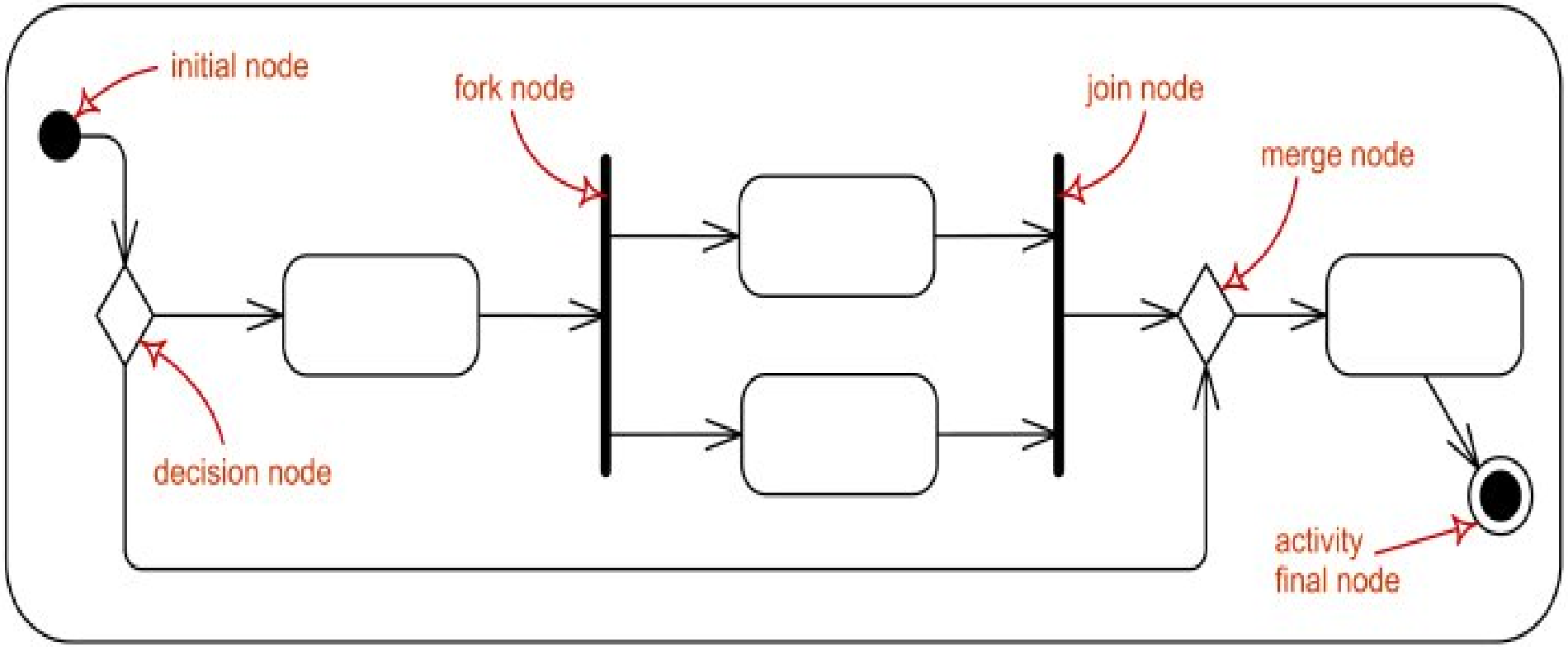
- Synchronization bars used to show joins in the workflow.
- ie. What activities must complete before processing may continue.
- All actions coming into the join must be completed before processing continues.

Fork



- Synchronization bars can also be used to show forks in the workflow.
- Actions on parallel flows beneath the folk can occur in any order or concurrently.

Activity control nodes overview.



Swim lanes

- *Swim lanes* may be used to partition an activity diagram.
- This facility allows activity diagrams to expand and show who has the responsibility for each activity in a process.

Swim lanes cont…

Eg:- Consider a consulting firm and the business process involved in meeting a new client;

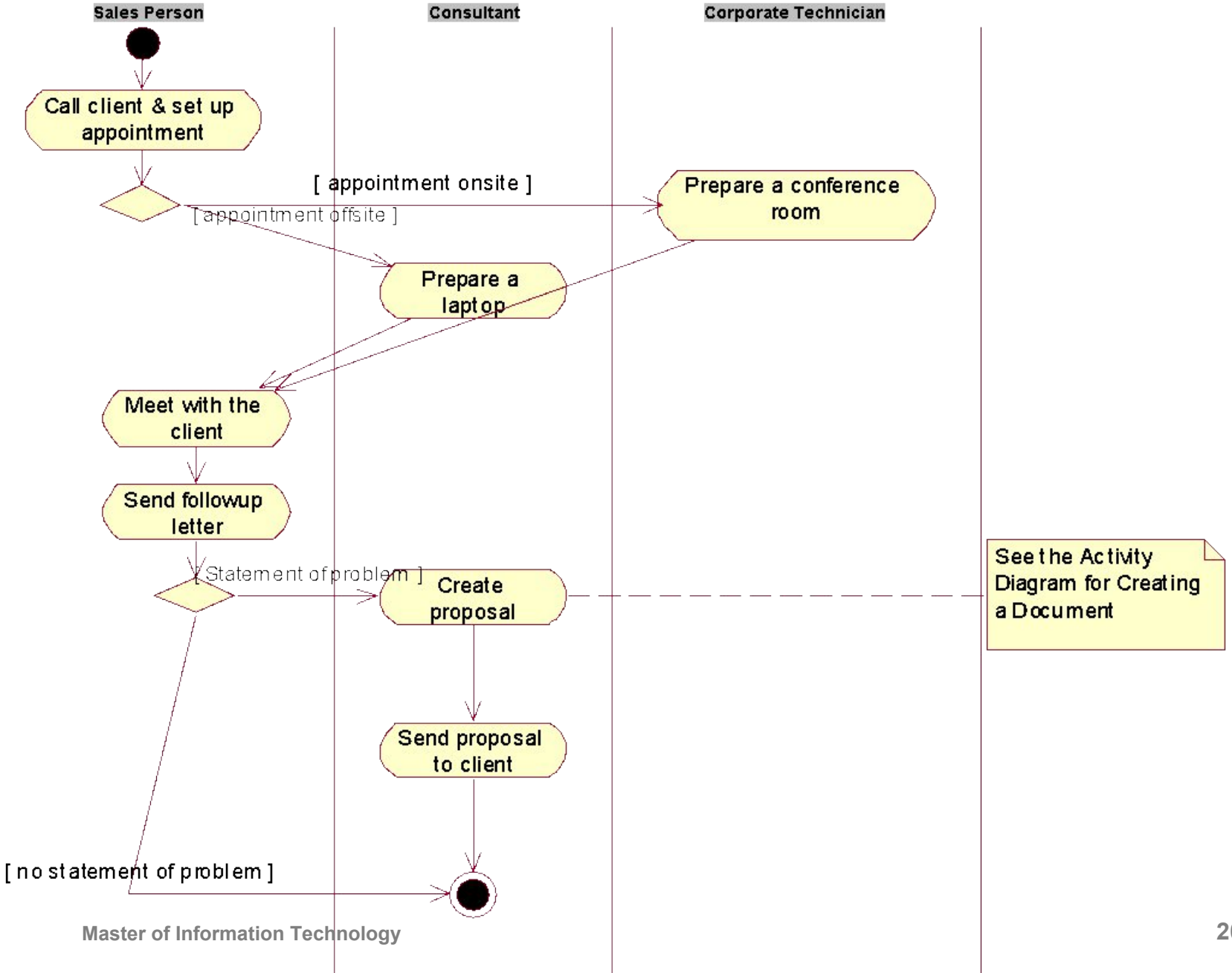
Following are the activities:

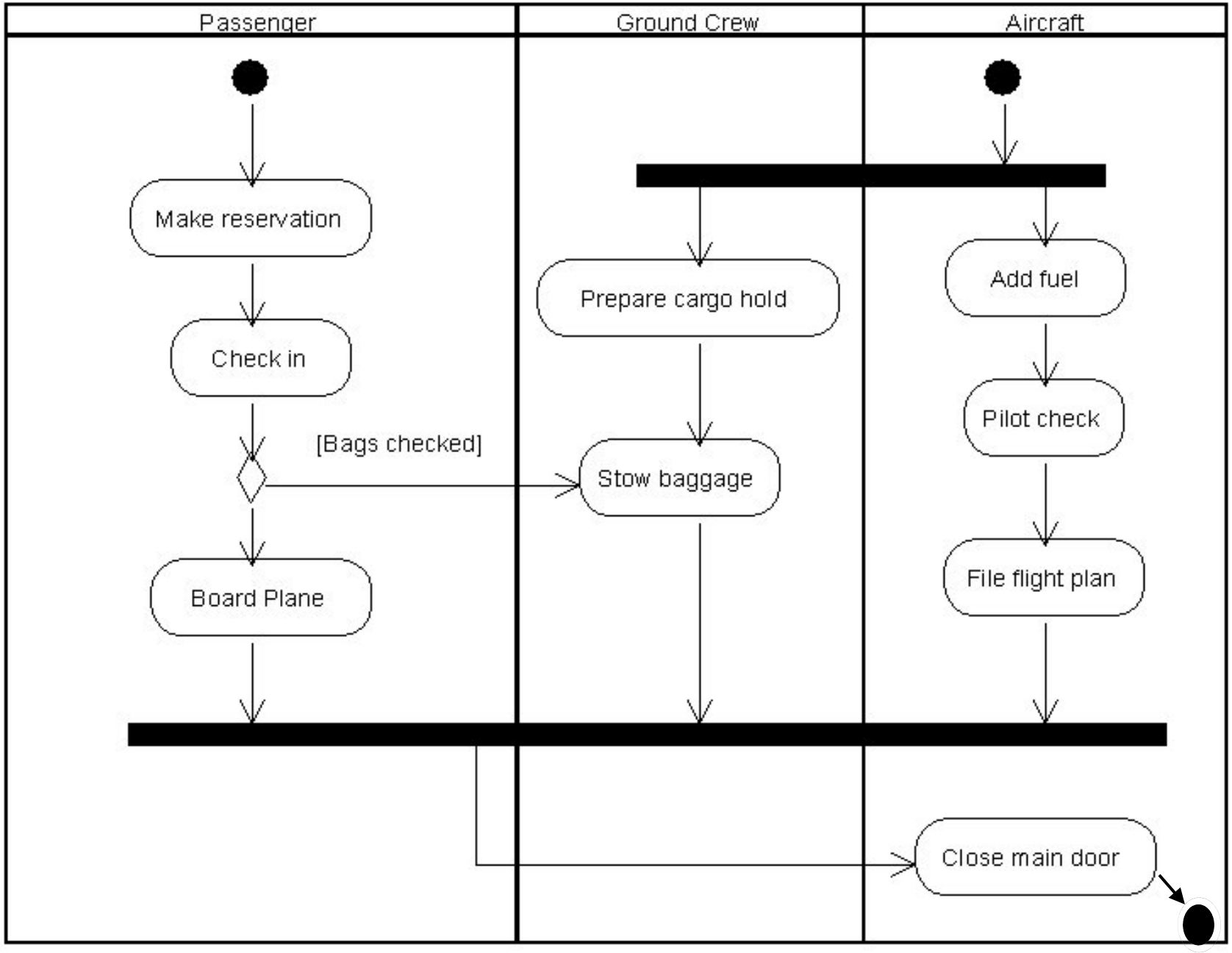
- A salesperson calls the client and sets up an appointment.
- If the appointment is onsite, corporate technicians prepare a conference room for a presentation.

Swim lanes cont…

- If the appointment is offsite, a consultant prepares a presentation on a laptop.
- The consultant and the salesperson meet with the client at the agreed-upon location and time.
- The salesperson follows up with a letter.
- If the meeting has resulted in a statement of a problem, the consultant creates a proposal and sends it to the client

An Activity Diagram



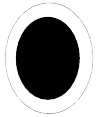

Passenger
Ground Crew
Aircraft
Make reservation
Check in
[Bags checked]
Board Plane
Prepare cargo hold
Stow baggage
Add fuel
Pilot check
File flight plan
Close main door

Activity Diagrams Cont…

Initial and Final Activities

- There are special symbols that are used to show the starting and final activities in a work flow.
- Starting activities are shown using a solid filled circle.

- The final activities are shown using a bull's eye.



Activity Diagram cont….

- Typically there is one starting activity for the workflow and
- There may be more than one ending activity (one for each alternate flow in the work flow)

Activity Diagram cont…

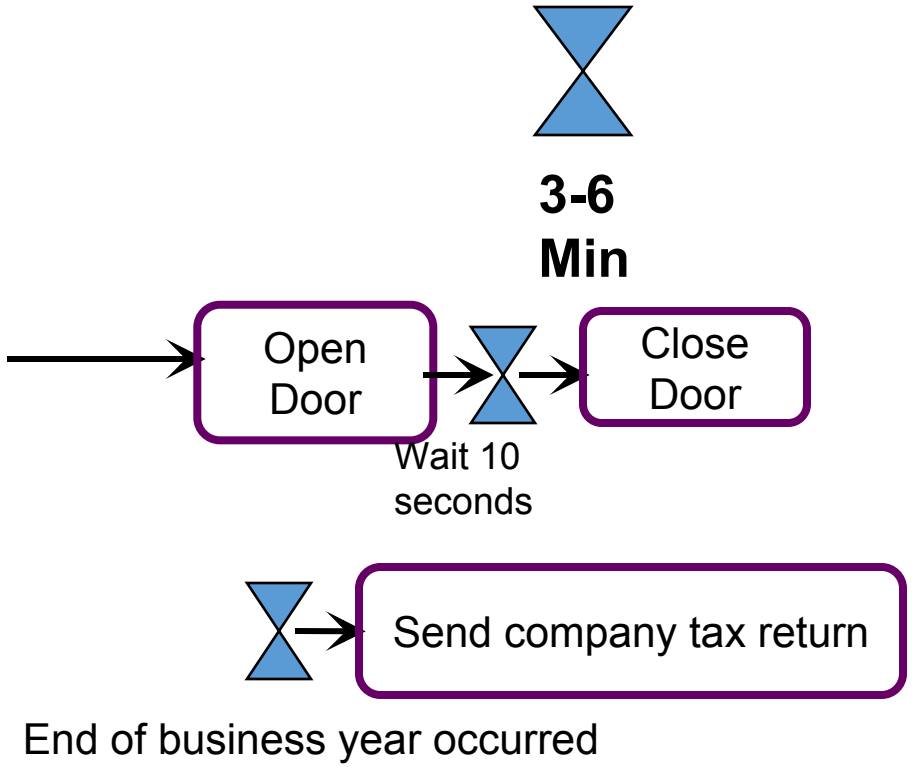
- Activity Diagrams are useful for communicating logic to programmers.
- Be careful however trying to use them to communicate logic to users.
- Non technical users may have trouble following them.
- You are better off using use case narratives with users.

New concepts included in UML 2.0

Passage of time =>

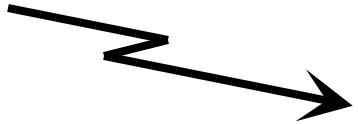
Eg. Accept time event

Eg. Action Node



New concepts included in UML 2.0

- Exception activity
  - Use to show an interrupting activity.

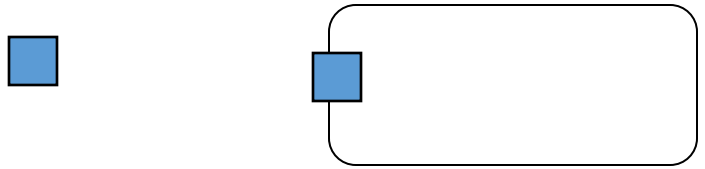


- Subactivity indicator
  - Action invokes another activity diagram

New concepts included in UML 2.0

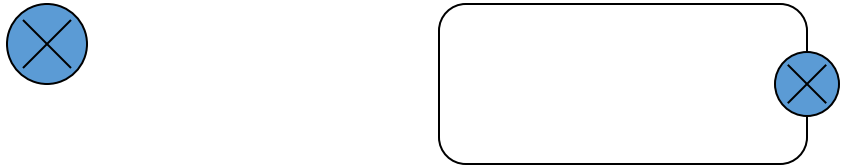
Input/output nodes ,

- Specifies an activity's Input and output nodes

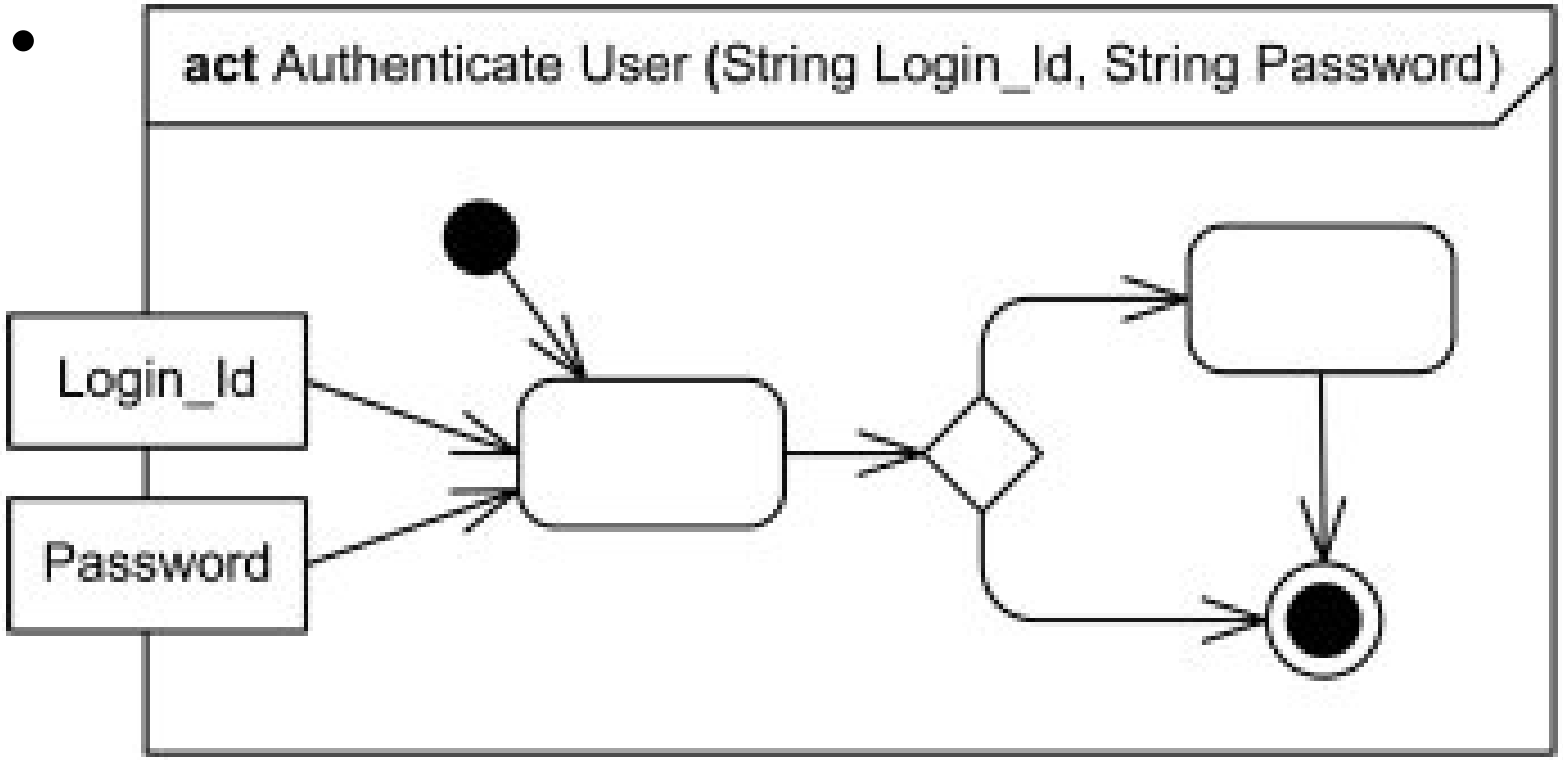


Flow final node

- Terminates a specific flow within an activity. Other flows are unaffected.

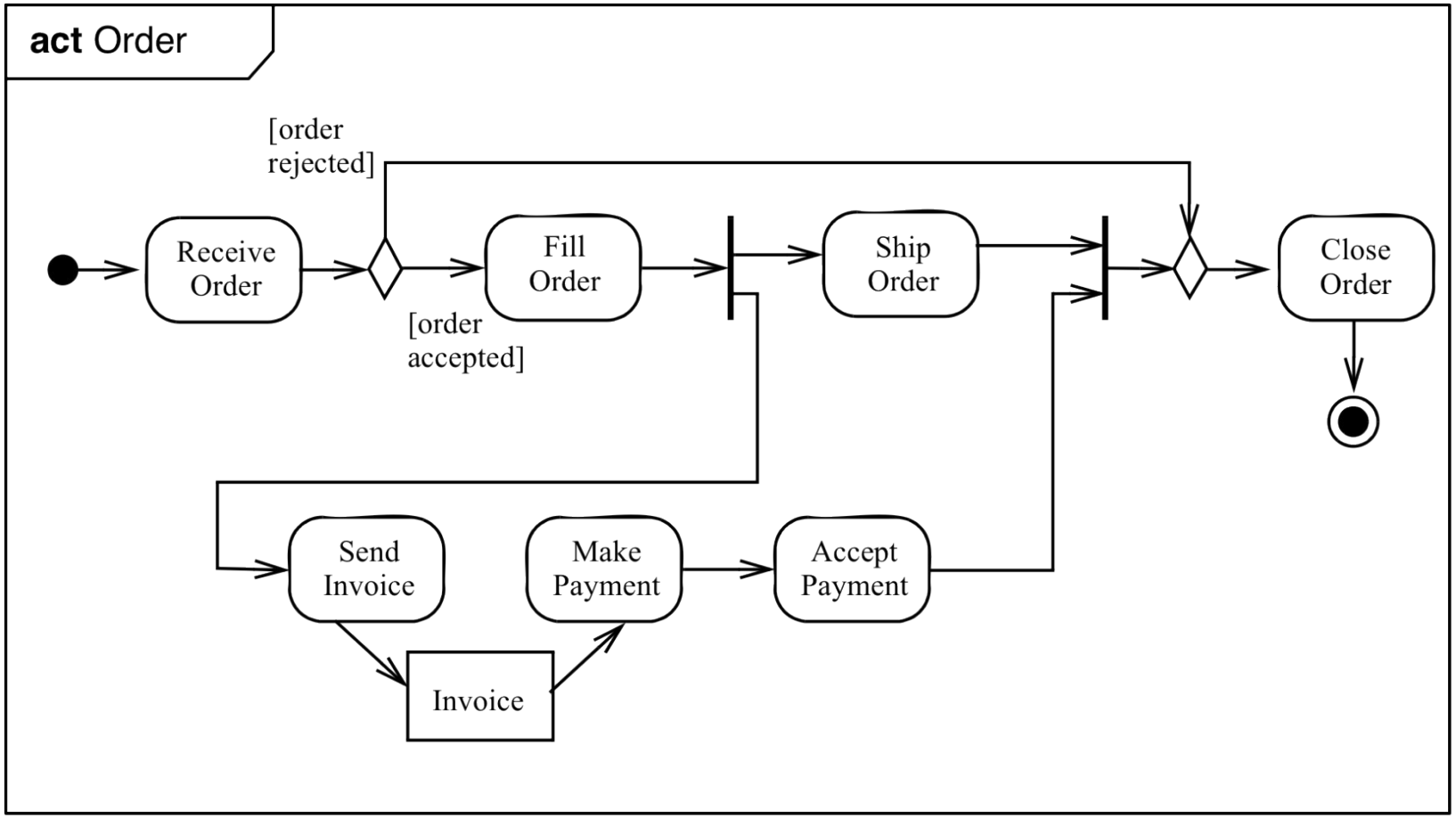


New concepts included in UML 2.0

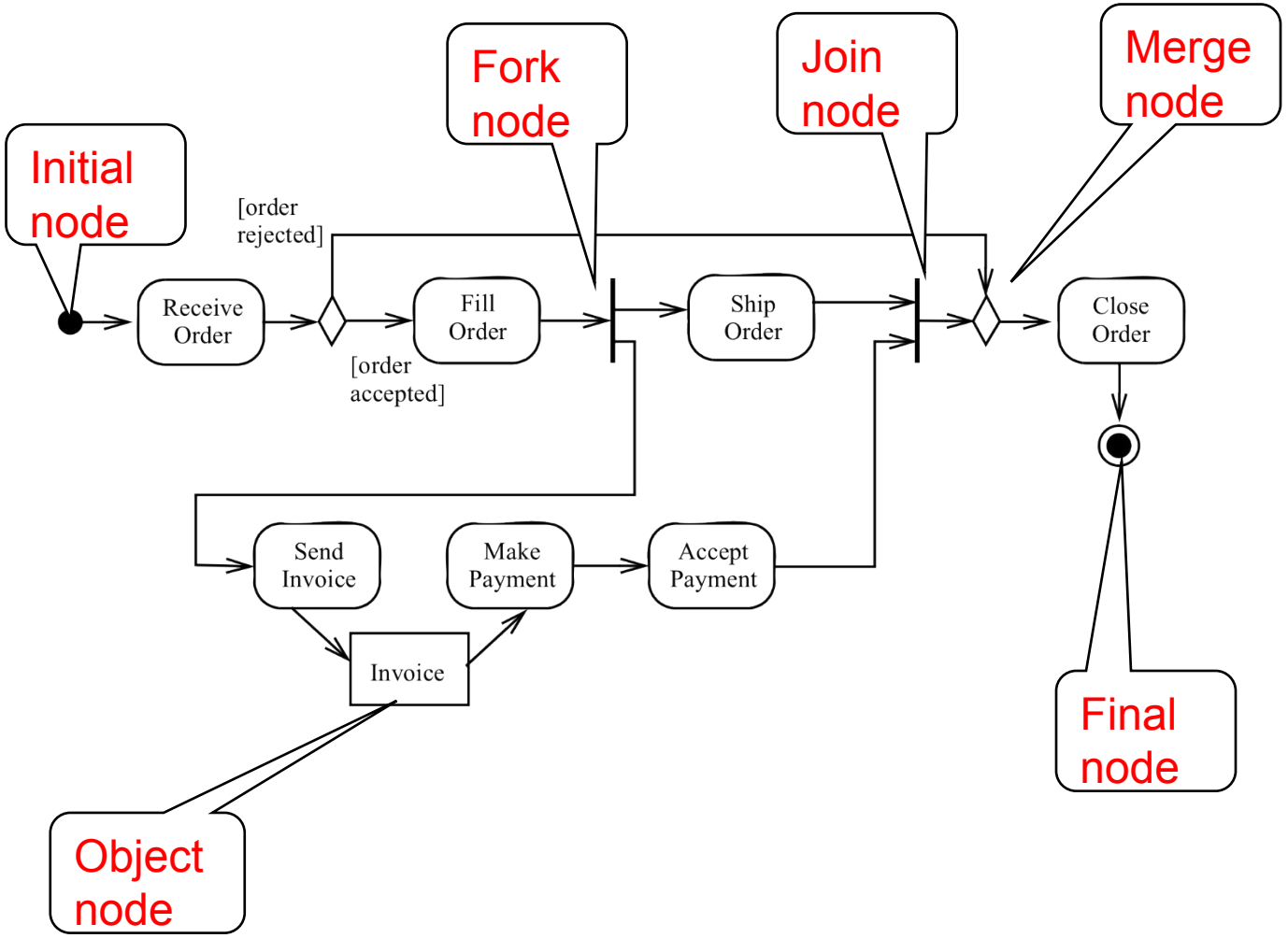
- 

Authenticate User activity frame with two parameters - Login Id and Password. (UML 2.4)

Example of a Activity Diagram with a Frame

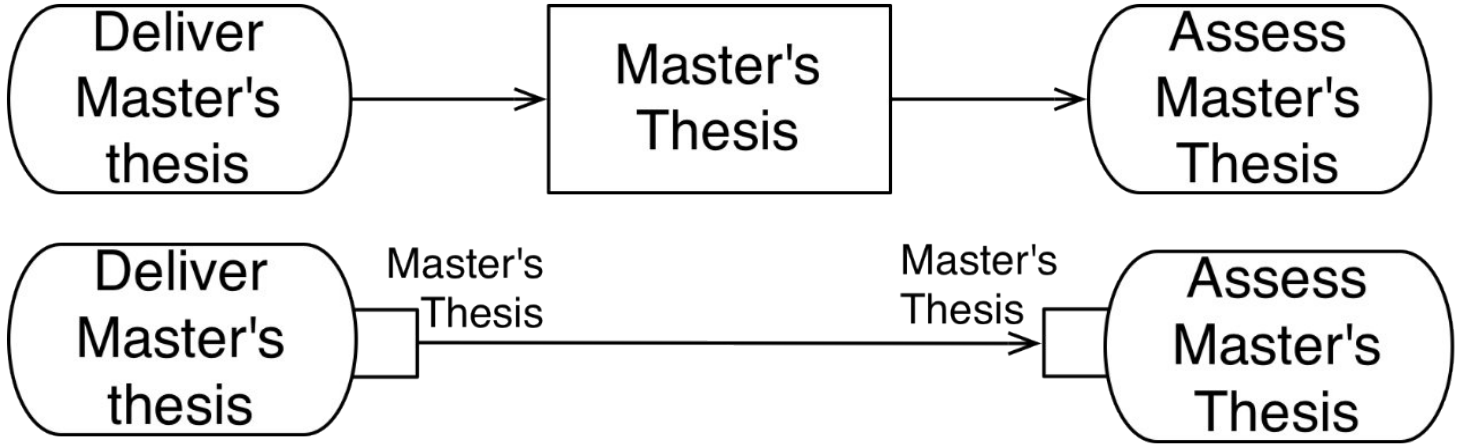


Activity Diagram Example



Activity Diagram: Pins

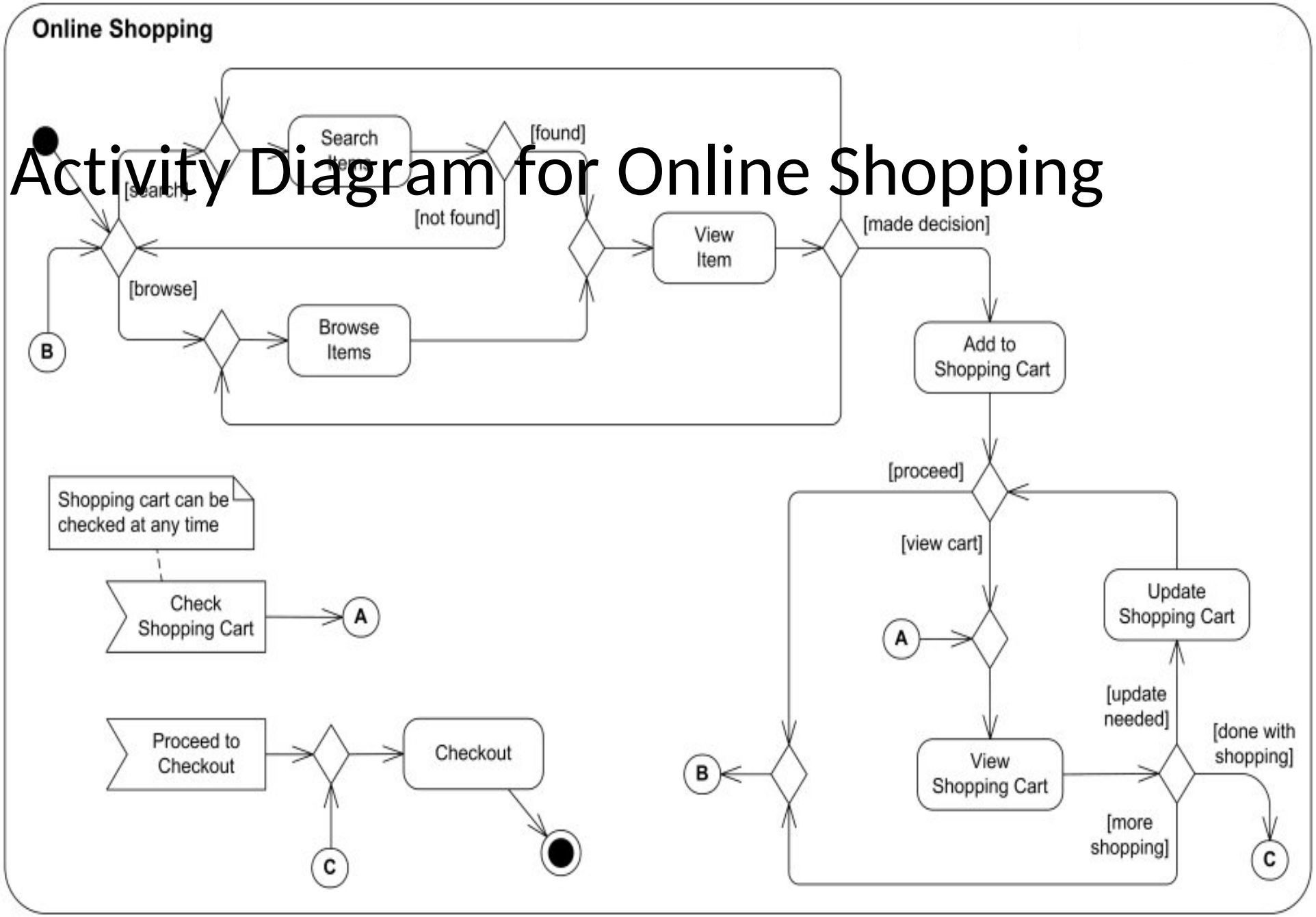
- Pin: Abbreviated notation for an object node
  - Different notations with same semantics
- Both notations define object flow in an activity



Example for an Activity Diagram for Online Shopping

- **Summary**: Online customer can browse or search items, view specific item, add it to shopping cart, view and update shopping cart, do checkout. User can view shopping cart at any time.

Activity Diagram for Online Shopping

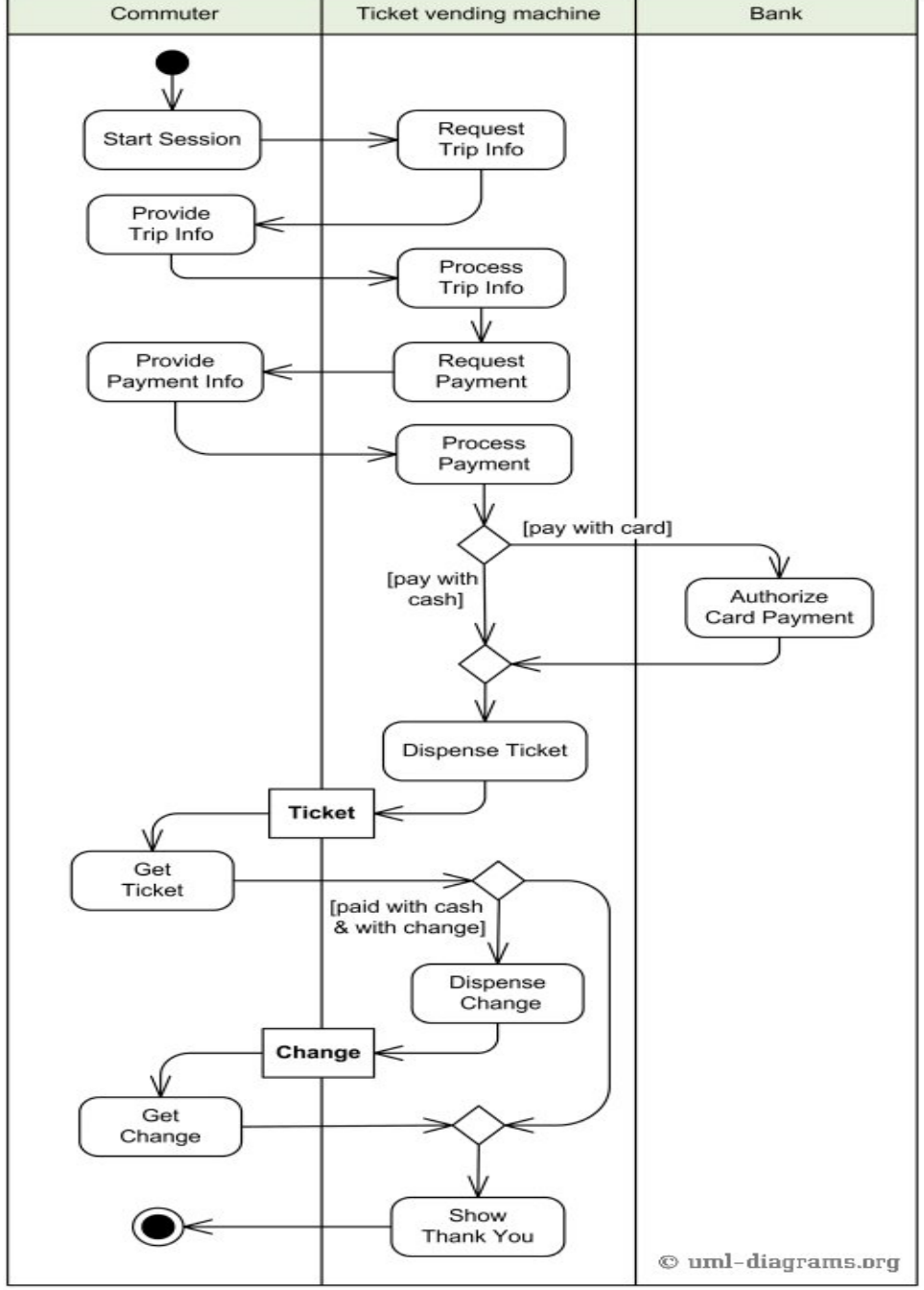


An example of UML activity diagram -

Purchase Ticket use case for a Ticket vending machine.

Summary : Activity is started by Commuter actor who needs to buy a ticket. Ticket vending machine will request trip information from Commuter. Based on the info machine will calculate payment due and request payment options. After payment is complete, ticket is dispensed to the Commuter.

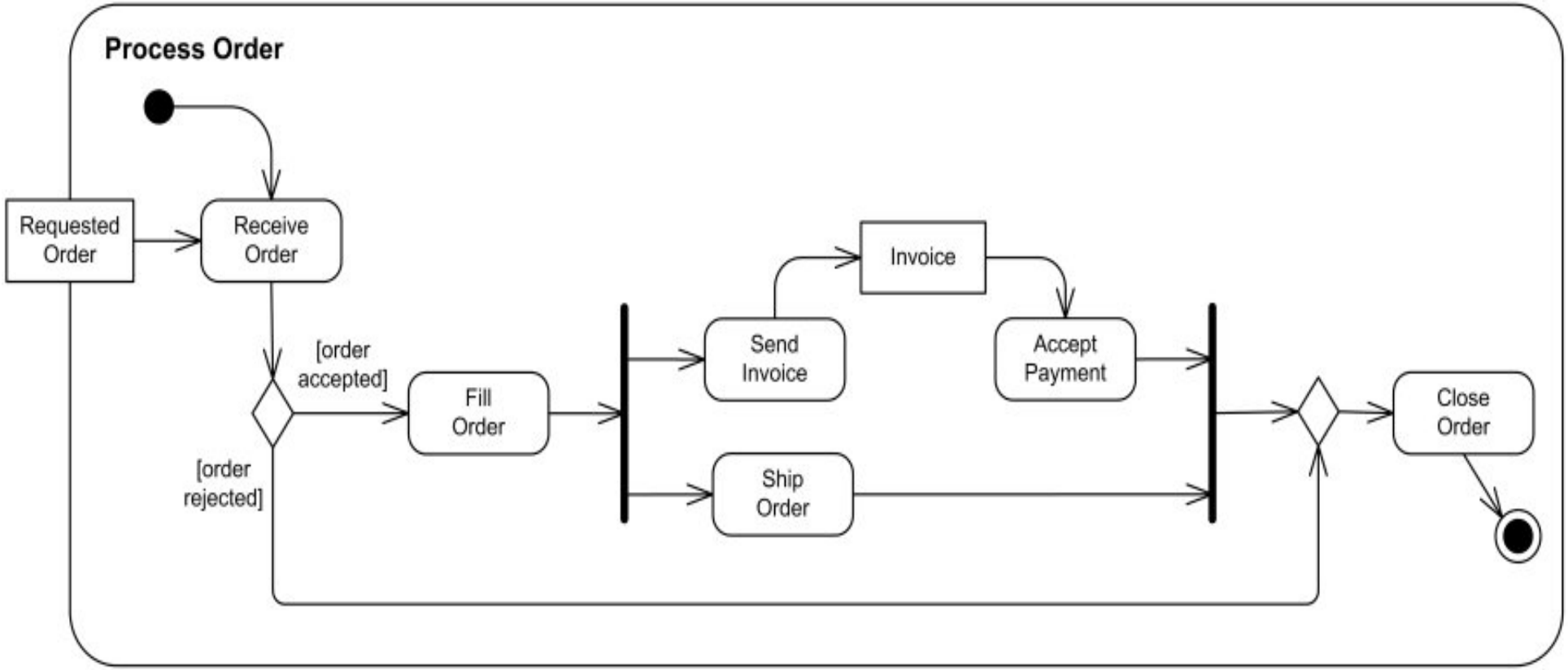
Purchase Ticket Vending Machine



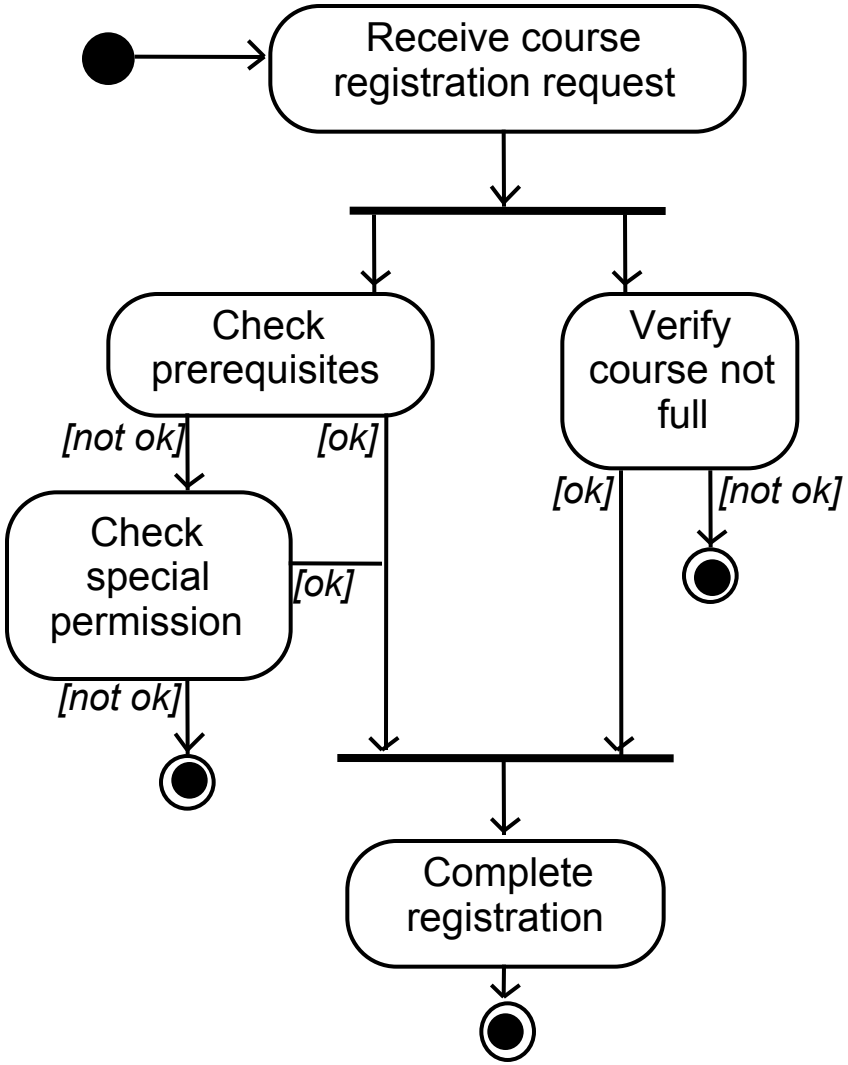
An example of business flow activity to process purchase order.

Summary : Requested order is input parameter of the activity. After order is accepted and all required information is filled in, payment is accepted and order is shipped. Note, that this business flow allows order shipment before invoice is sent or payment is confirmed.

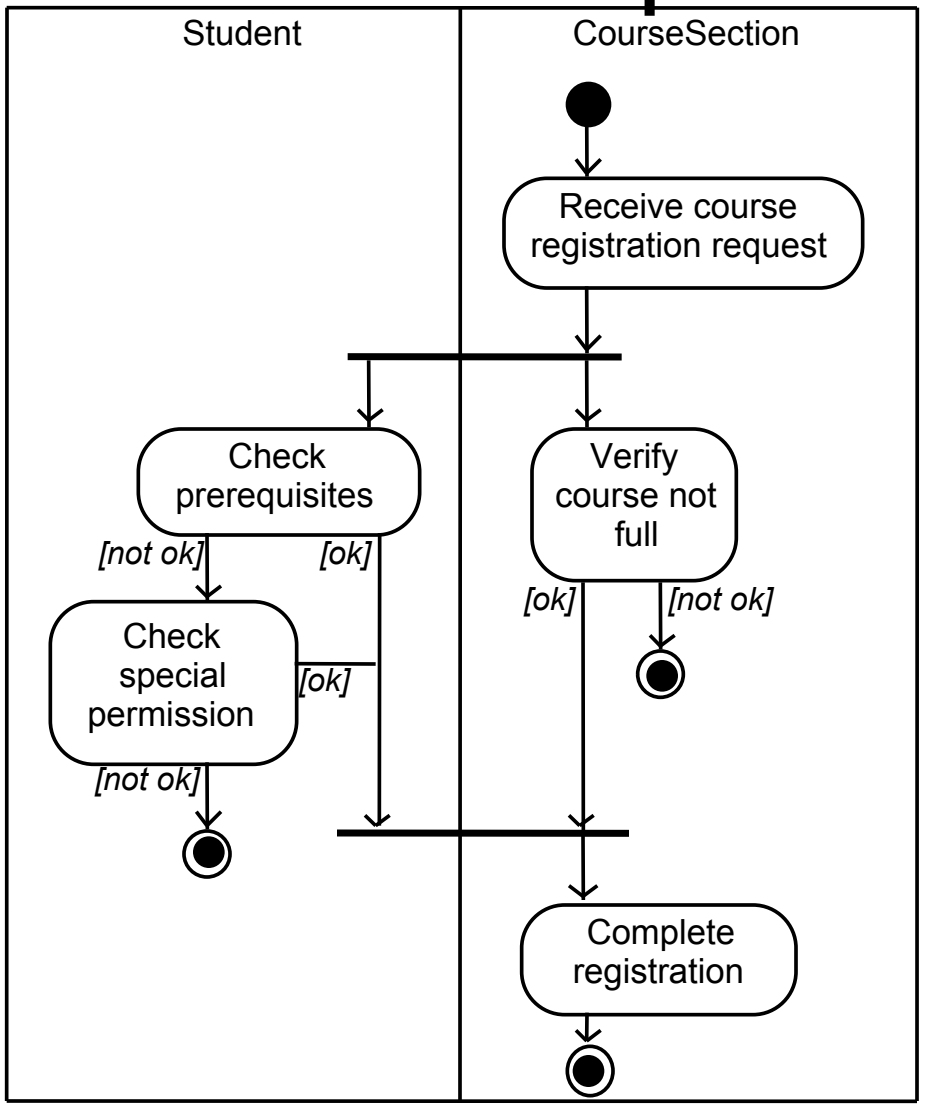
An example of business flow activity to process purchase order.



Activity diagrams – more examples



Activity diagrams – an example with swimlanes



Representing concurrency

- Concurrency is shown using forks, joins and rendezvous.
  - A *fork* has one incoming transition and multiple outgoing transitions.
    - The execution splits into two concurrent threads.
  - A *rendezvous* has multiple incoming and multiple outgoing transitions.
    - Once all the incoming transitions occur all the outgoing transitions may occur.

Representing concurrency

- A *join* has <u>multiple</u> incoming transitions and <u>one</u> outgoing transition.
  - The outgoing transition will be taken when all incoming transitions have occurred.
  - The incoming transitions must be triggered in separate threads.
  - If one incoming transition occurs, a wait condition occurs at the join until the other transitions occur.

Q & A